



**Generation Interconnection
Impact Study Report
for
Queue Project AE2-188
MITCHELL DP-MOUNTAIN RUN 115 KV
40 MW Capacity / 40 MW Energy**

February, 2020

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

3 General

The Interconnection Customer (IC) has proposed a new Storage generating facility located in Culpeper County, Virginia. The installed facilities will have a total capability of 40 MW with 40 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 31, 2020. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-188
Project Name	MITCHELL DP-MOUNTAIN RUN 115 KV
Interconnection Customer	Invenergy Solar Development North America LLC
State	Virginia
County	Culpeper
Transmission Owner	Dominion
MFO	40
MWE	40
MWC	40
Fuel	Storage
Basecase Study Year	2022

3.1 Point of Interconnection

AE2-188 will interconnect with the Dominion transmission system through the AC2-102 115 kV interconnection switchyard which will intersect the Mountain Run to Mitchell DP 115 kV line. The AE2-188 project will be sharing Attachment Facilities with the AC2-102 project. Attachment 1 shows a one-line diagram of the proposed interconnection.

3.2 Cost Summary¹

The AE2-188 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$ TBD in Facilities Study
Total Costs	\$ 0

In addition, the AE2-188 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$5,905,517

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less

¹ Note that the AE2-188 project intends to share Attachment Facilities with the AC2-102 project. Should the AC2-102 project drop from the Interconnection Queue, the AE2-188 project will be responsible to assume the Attachment Facilities, Direct Connection and Non-Direction costs identified in the AC2-102 System Impact Study report for connection to the Dominion system.

than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

4 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AE2-188 was evaluated as a 40.0 MW Capacity (40.0 MW energy) injection at the AE2-188 500 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2022 AE2 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically in Planning Studies NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

The required Attachment Facilities, Direct Connection and Non-Direct Connection work for the interconnection of the AE2-188 generation project to the Dominion Transmission System is detailed in the following sections. The associated one-line with the generation project attachment facilities and primary direct and non-direct connection are shown in Attachment 1.

Note that the ITO findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in a future study phases. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

5 Non-Direct Connection Cost Estimate

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

6 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

7 Transmission Owner Analysis

7.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2022 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

7.2 Short Circuit Analysis

PJM performed a short circuit analysis and the results were verified by Dominion. The connection of AE2-188 project to the system does not result in any newly overdutied circuit breakers on the Dominion transmission system and does not have a significant fault current contribution to existing overdutied circuit breakers

7.3 Stability Analysis

PJM will complete a dynamic stability analysis as part of the Facilities Study. The results of this analysis will be reviewed by Dominion. Should stability concerns be identified in PJM's study, Dominion will develop appropriate system reinforcement(s) and included the estimated cost of any reinforcement(s) in Dominion's System Impact Study report.

8 Interconnection Customer Requirements

8.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary

Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

8.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion's "Dominion's Facility Interconnection Requirements" document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

8.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electrictransmission-access>.

10 Network Impacts

The Queue Project AE2-188 was evaluated as a 40 MW (Capacity 40 MW) injection at the AC1-102 substation which will tap the Mitchel DP and Mountain Run 115 kV line in the Dominion area. Project AE2-188 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-188 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

11 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

12 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
1946965	313703	3UNIONVILLE	115.0	DVP	314764	3LOCSTG	115.0	DVP	1	DVP_P4-2: 2T70	breaker	235.0	93.81	103.3	AC	22.84
1948221	313703	3UNIONVILLE	115.0	DVP	314764	3LOCSTG	115.0	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	235.0	99.37	109.09	AC	23.35
10465287	313703	3UNIONVILLE	115.0	DVP	314764	3LOCSTG	115.0	DVP	1	DVP_P7-1: LN 70-2153-A-A	tower	235.0	95.62	105.35	AC	23.35
1948226	314778	3PAY TAP	115.0	DVP	314755	3SPOTSYL	115.0	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	260.0	92.24	100.9	AC	23.35

13 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
1946646	314755	3SPOTSYL	115.0	DVP	314779	3TDTAVRN	115.0	DVP	1	DVP_P4-2: 573T594	breaker	260.0	140.37	144.35	AC	10.35
1946739	314775	3NIVER	115.0	DVP	314135	3CHANCE	115.0	DVP	1	DVP_P4-2: 573T594	breaker	260.0	127.46	131.4	AC	10.35
1946711	314779	3TDTAVRN	115.0	DVP	314775	3NIVER	115.0	DVP	1	DVP_P4-2: 573T594	breaker	260.0	131.35	135.3	AC	10.35
1946893	926000	AC1-076 TAP	115.0	DVP	314778	3PAY TAP	115.0	DVP	1	DVP_P4-2: 2T70	breaker	260.0	100.4	108.9	AC	22.84
1948188	926000	AC1-076 TAP	115.0	DVP	314778	3PAY TAP	115.0	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	260.0	105.67	114.38	AC	23.35

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
10465226	926000	AC1-076 TAP	115.0	DVP	314778	3PAY TAP	115.0	DVP	1	DVP_P7-1: LN 70-2153-A-A	tower	260.0	102.29	111.01	AC	23.35

14 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
1947797	313703	3UNIONVILLE	115.0	DVP	314764	3LOCST G	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	191.76	114.61	126.23	AC	22.84
1947888	314764	3LOCST G	115.0	DVP	926000	AC1-076 TAP	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	212.44	102.7	113.2	AC	22.84
1947500	314768	3MITCHEL	115.0	DVP	314815	3OAK GRE	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	217.14	138.01	155.83	AC	40.0
1947764	314778	3PAY TAP	115.0	DVP	314755	3SPOTSYL	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	212.44	106.16	116.49	AC	22.84
1947909	314815	3OAK GRE	115.0	DVP	313703	3UNIONVILLE	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	224.66	100.94	110.87	AC	22.84
1947980	925670	AC1-043 TAP	115.0	DVP	314345	3MT RUN	115.0	DVP	1	DVP_P1-2: LN 2-D	operation	331.82	91.87	103.76	AC	40.0
1947675	926000	AC1-076 TAP	115.0	DVP	314778	3PAY TAP	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	212.44	122.55	132.96	AC	22.84
1947978	926480	AC1-120 TAP	115.0	DVP	314768	3MITCHEL	115.0	DVP	1	DVP_P1-2: LN 2-A	operation	331.82	91.78	103.46	AC	40.0

15 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-188	Upgrade Number
1948226	2	3PAY TAP 115.0 kV - 3SPOTSYL 115.0 kV Ckt 1	n6375: Rebuild 2.83 miles of 115 kV Line 153 from Paytes Tap to Spotsylvania with 768 ACSS. Project Type : FAC Cost : \$3.679 Million Time Estimate : 30-36 months New Rating: Rate A: 393 Rate B: 393 Rate C: 452 Next limiting element: Line SW/Wavetraps Upgrade for 'DVP_P7-1: LN 70-2153-B-A' AE2-188 is the driver for this upgrade.	\$3,679,000	\$3,679,000	n6375
1946739	4	3NI RVER 115.0 kV - 3CHANCE 115.0 kV Ckt 1	n6372: Rebuild 1.34 miles of 115 kV Line 198 from Ni River DP to Chancellor with 768 ACSS. Project Type : FAC Cost : \$1.742 Million Time Estimate : 30-36 months New Rating: Rate A: 393 Rate B: 393 Rate C: 452 Next limiting element: Wavetraps/Line SW Queue Project AE2-188 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE2-188 could receive cost allocation. Note 2: Although Queue Project AE2-188 may not have cost responsibility for this upgrade, Queue Project AE2-188 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-188 comes into service prior to completion of the upgrade, Queue Project AE2-188 will need an interim study	\$1,742,000	\$0	n6372

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE2-188	Upgrade Number															
1948221,1946965,10465287	1	3UNIONVILLE 115.0 kV - 3LOCST G 115.0 kV Ckt 1	n6374: Rebuild 0.06 miles of 115 kV Line 153 from Unionville DP to Locust Grove DP with 768 ACSS. Project Type : FAC Cost : \$0.5 Million Time Estimate : 30-36 months New Rating: Rate A: 261 Rate B: 261 Rate C: 301 Next limiting element: Conductor. Upgrade for 'DVP_P7-1: LN 70-2153-B-A' AE2-188 is the driver for this upgrade.	\$500,000	\$500,000	n6374															
1946646	3	3SPOTSYL 115.0 kV - 3TDTAVRN 115.0 kV Ckt 1	n6371: Rebuild 7.46 miles of 115 kV Line 198 from Spotsylvania to Todds Tavern DP with 768 ACSS. Project Type : FAC Cost : \$9.698 Million Time Estimate: 30-36 months New Rating: Rate A: 393 Rate B: 393 Rate C: 452 Next limiting element: Wavetraps/Line SW/Leads The cost allocation table is below: <table><tr><td>Queue</td><td>MW contribution</td><td>Cost \$9.698M</td><td>Contingency Name</td><td>Contingency Type</td></tr><tr><td>AE1 Queue</td><td>97.90</td><td>\$8,770,755</td><td></td><td></td></tr><tr><td>AE2-188</td><td>10.35</td><td>\$927,245</td><td>'DVP_P4-2: 573T594'</td><td>Breaker</td></tr></table>	Queue	MW contribution	Cost \$9.698M	Contingency Name	Contingency Type	AE1 Queue	97.90	\$8,770,755			AE2-188	10.35	\$927,245	'DVP_P4-2: 573T594'	Breaker	\$9,698,000	\$927,245	n6371
Queue	MW contribution	Cost \$9.698M	Contingency Name	Contingency Type																	
AE1 Queue	97.90	\$8,770,755																			
AE2-188	10.35	\$927,245	'DVP_P4-2: 573T594'	Breaker																	

ID	Index	Facility	Upgrade Description						Cost	Cost Allocated to AE2-188	Upgrade Number
1946711	5	3TDTAVRN 115.0 kV - 3NI RVER 115.0 kV Ckt 1	n6373: Rebuild 5.07 miles of 115 kV Line 198 from Todds Tavern DP to Ni River DP with 768 ACSS.						\$6,591,000	\$799,272	n6373
			Project Type : FAC								
			Cost : \$6.591 Million								
			Time Estimate : 30-36 months								
			New Rating: Rate A: 393 Rate B: 393 Rate C: 452								
Next limiting element: Line SW											
The cost allocation table is below:											
			Queue	MW contri butio n	Perc enta ge of Cost	Cost(\$6.59 1M)	Contingency Name	Con ger Typ			
			Unkno wn	74.99	87.8 7%	\$5,791,728					
			AE2- 188	10.35	12.1 3%	\$799,272	'DVP_P4-2: 573T594'	Brea r			
10465226,1948188,1946893	6	AC1-076 TAP 115.0 kV - 3PAY TAP 115.0 kV Ckt 1	n5612: Rebuild 2.8 miles of 115 kV Line 153 from AC1-076 Tap to Paytes Tap with 768 ACSS.						\$3,653,000	\$0	n5612
			Project Type : FAC								
			Cost : \$3.653 Million								
			Time Estimate : 30-36 months								
			New Rating : Rate A: 393 Rate B: 393 Rate C: 452								
Queue Project AE2-188 presently does not receive cost allocation for this upgrade.											
Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AE2-188 could receive cost allocation.											
Note 2: Although Queue Project AE2-188 may not have cost responsibility for this upgrade, Queue Project AE2-188 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-188 comes into service prior to completion of the upgrade, Queue Project AE2-188 will need an interim study											
			TOTAL COST						\$25,863,000	\$5,905,517	

16 Flow Gate Details

The following indices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

16.1 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P1-2: LN 2-D	CONTINGENCY 'DVP_P1-2: LN 2-D' OPEN BRANCH FROM BUS 926480 TO BUS 314768 CKT 1 /* AC1-120 TAP 115.00 - 3MITCHEL 115.00 OPEN BRANCH FROM BUS 314768 TO BUS 314815 CKT 1 /* 3MITCHEL 115.00 - 3OAK GRE 115.00 OPEN BUS 314768 /* ISLAND: 3MITCHEL 115.00 END
DVP_P4-2: 573T594	CONTINGENCY 'DVP_P4-2: 573T594' /* SPOTSYLVANIA 500 KV OPEN BRANCH FROM BUS 314918 TO BUS 314934 CKT 1 /* 8NO ANNA 500.00 - 8SPOTSYL 500.00 OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 END
DVP_P1-2: LN 2-A	CONTINGENCY 'DVP_P1-2: LN 2-A' OPEN BRANCH FROM BUS 314345 TO BUS 925670 CKT 1 /* 3MT RUN 115.00 - AC1-043 TAP 115.00 END
DVP_P7-1: LN 70-2153-B-A	CONTINGENCY 'DVP_P7-1: LN 70-2153-B-A' OPEN BRANCH FROM BUS 926610 TO BUS 314743 CKT 1 /* AC1-143 TAP 115.00 - 3BRANDY 115.00 OPEN BRANCH FROM BUS 314345 TO BUS 314743 CKT 1 /* 3MT RUN 115.00 - 3BRANDY 115.00 OPEN BRANCH FROM BUS 314345 TO BUS 314753 CKT 1 /* 3MT RUN 115.00 - 3CULPEPR 115.00 OPEN BRANCH FROM BUS 314753 TO BUS 314873 CKT 1 /* 3CULPEPR 115.00 - 3CULPE_1 115.00 OPEN BUS 314743 /* ISLAND: 3BRANDY 115.00 OPEN BUS 314753 /* ISLAND: 3CULPEPR 115.00 OPEN BUS 314873 /* ISLAND: 3CULPE_1 115.00 OPEN BRANCH FROM BUS 314080 TO BUS 939220 CKT 1 /* 6REMNGTN 230.00 - AE1-153 TAP 230.00 END

Contingency Name	Contingency Definition
DVP_P4-2: 2T70	CONTINGENCY 'DVP_P4-2: 2T70' /* MOUNTAIN RUN 115 KV OPEN BRANCH FROM BUS 314345 TO BUS 925670 CKT 1 /* MT. RUN 115.00 - AC1-043 TAP 115.00 /* CONTINGENCY LINE ADDED FOR AE1 BUILD OPEN BRANCH FROM BUS 926610 TO BUS 314743 CKT 1 /* AC1-143 TAP 115.00 115.00 - 3BRANDY 115.00 OPEN BRANCH FROM BUS 314345 TO BUS 314743 CKT 1 /* 3MT RUN 115.00 - 3BRANDY 115.00 OPEN BRANCH FROM BUS 314345 TO BUS 314753 CKT 1 /* 3MT RUN 115.00 - 3CULPEPR 115.00 OPEN BRANCH FROM BUS 314753 TO BUS 314873 CKT 1 /* 3CULPEPR 115.00 - 3CULPE_1 115.00 OPEN BUS 314743 /* ISLAND: 3BRANDY 115.00 OPEN BUS 314753 /* ISLAND: 3CULPEPR 115.00 OPEN BUS 314873 /* ISLAND: 3CULPE_1 115.00 END
DVP_P7-1: LN 70-2153-A-A	CONTINGENCY 'DVP_P7-1: LN 70-2153-A-A' OPEN BRANCH FROM BUS 314078 TO BUS 926610 CKT 1 /* 3REMNGTN 115.00 - AC1-143 TAP 115.00 OPEN BRANCH FROM BUS 314080 TO BUS 939220 CKT 1 /* 6REMNGTN 230.00 - AE1-153 TAP 230.00 END

16.2 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1948221	313703	3UNIONVILLE	DVP	314764	3LOCST G	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	235.0	99.37	109.09	AC	23.35

Bus #	Bus	MW Impact
925671	AC1-043 C	22.1859
925672	AC1-043 E	36.1981
926481	AC1-120 C O1	23.1201
926482	AC1-120 E O1	11.9103
926501	AC1-121 C O1	7.9402
926502	AC1-121 E O1	3.7366
932781	AC2-102 C	17.7487
932782	AC2-102 E	28.9585
934861	AD1-115 C	11.0930
934862	AD1-115 E	18.0990
939221	AE1-153 C O1	6.7813
939222	AE1-153 E O1	4.5209
941831	AE2-188	23.3536
LGEE	LGEE	0.1019
CIN	CIN	0.3430
CPLE	CPLE	0.1475
IPL	IPL	0.2174
G-007	G-007	0.2341
CBM-W2	CBM-W2	3.8200
CBM-W1	CBM-W1	0.7295
MECS	MECS	0.3430
WEC	WEC	0.0884
O-066	O-066	1.4806
CBM-S2	CBM-S2	0.3657
CARR	CARR	0.0684
CBM-S1	CBM-S1	0.5699
MEC	MEC	0.6699
RENSSELAER	RENSSELAER	0.0540

16.3 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1948226	314778	3PAY TAP	DVP	314755	3SPOTSYL	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	260.0	92.24	100.9	AC	23.35

Bus #	Bus	MW Impact
925671	AC1-043 C	22.1859
925672	AC1-043 E	36.1981
926001	AC1-076 C	17.0287
926002	AC1-076 E	15.9554
926481	AC1-120 C O1	23.1201
926482	AC1-120 E O1	11.9103
926501	AC1-121 C O1	7.9402
926502	AC1-121 E O1	3.7366
932781	AC2-102 C	17.7487
932782	AC2-102 E	28.9585
934861	AD1-115 C	11.0930
934862	AD1-115 E	18.0990
939221	AE1-153 C O1	6.7813
939222	AE1-153 E O1	4.5209
941381	AE2-134	11.7340
941831	AE2-188	23.3536
LGEE	LGEE	0.1019
CIN	CIN	0.3430
CPLE	CPLE	0.1475
IPL	IPL	0.2174
G-007	G-007	0.2341
CBM-W2	CBM-W2	3.8200
CBM-W1	CBM-W1	0.7295
MECS	MECS	0.3430
WEC	WEC	0.0884
O-066	O-066	1.4806
CBM-S2	CBM-S2	0.3657
CARR	CARR	0.0684
CBM-S1	CBM-S1	0.5699
MEC	MEC	0.6699
RENSSELAER	RENSSELAER	0.0540

16.4 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1946646	314755	3SPOTSYL	DVP	314779	3TDTAVRN	DVP	1	DVP_P4-2: 573T594	breaker	260.0	140.37	144.35	AC	10.35

Bus #	Bus	MW Impact
925671	AC1-043 C	8.7138
925672	AC1-043 E	14.2172
926001	AC1-076 C	10.3906
926002	AC1-076 E	9.7357
926481	AC1-120 C O1	10.6461
926482	AC1-120 E O1	5.4843
926501	AC1-121 C O1	3.6562
926502	AC1-121 E O1	1.7206
926611	AC1-143 C O1	3.2705
926612	AC1-143 E O1	1.4924
926731	AC1-158 C	195.8649
926732	AC1-158 E	85.9551
932781	AC2-102 C	7.8651
932782	AC2-102 E	12.8325
934861	AD1-115 C	4.3569
934862	AD1-115 E	7.1086
941381	AE2-134	7.1599
941831	AE2-188	10.3488
LGEE	LGEE	0.0211
CIN	CIN	0.0710
IPL	IPL	0.0457
G-007	G-007	0.0431
CBM-W2	CBM-W2	0.4716
CBM-W1	CBM-W1	0.1563
MECS	MECS	0.0980
WEC	WEC	0.0180
O-066	O-066	0.2759
CARR	CARR	0.0110
CBM-S1	CBM-S1	0.0580
HAMLET	HAMLET	0.0265
MEC	MEC	0.1155
RENSSELAER	RENSSELAER	0.0090
CATAWBA	CATAWBA	0.0070

16.5 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1946739	314775	3NI RVER	DVP	314135	3CHANCE	DVP	1	DVP_P4-2: 573T594	breaker	260.0	127.46	131.4	AC	10.35

Bus #	Bus	MW Impact
925671	AC1-043 C	8.7138
925672	AC1-043 E	14.2172
926001	AC1-076 C	10.3906
926002	AC1-076 E	9.7357
926481	AC1-120 C O1	10.6461
926482	AC1-120 E O1	5.4843
926501	AC1-121 C O1	3.6562
926502	AC1-121 E O1	1.7206
926611	AC1-143 C O1	3.2705
926612	AC1-143 E O1	1.4924
926731	AC1-158 C	195.8649
926732	AC1-158 E	85.9551
932781	AC2-102 C	7.8651
932782	AC2-102 E	12.8325
934861	AD1-115 C	4.3569
934862	AD1-115 E	7.1086
941381	AE2-134	7.1599
941831	AE2-188	10.3488
LGEE	LGEE	0.0211
CIN	CIN	0.0710
IPL	IPL	0.0457
G-007	G-007	0.0431
CBM-W2	CBM-W2	0.4716
CBM-W1	CBM-W1	0.1563
MECS	MECS	0.0980
WEC	WEC	0.0180
O-066	O-066	0.2759
CARR	CARR	0.0110
CBM-S1	CBM-S1	0.0580
HAMLET	HAMLET	0.0265
MEC	MEC	0.1155
RENSSELAER	RENSSELAER	0.0090
CATAWBA	CATAWBA	0.0070

16.6 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1946711	314779	3TDTAVRN	DVP	314775	3NI RVER	DVP	1	DVP_P4-2: 573T594	breaker	260.0	131.35	135.3	AC	10.35

Bus #	Bus	MW Impact
925671	AC1-043 C	8.7138
925672	AC1-043 E	14.2172
926001	AC1-076 C	10.3906
926002	AC1-076 E	9.7357
926481	AC1-120 C O1	10.6461
926482	AC1-120 E O1	5.4843
926501	AC1-121 C O1	3.6562
926502	AC1-121 E O1	1.7206
926611	AC1-143 C O1	3.2705
926612	AC1-143 E O1	1.4924
926731	AC1-158 C	195.8649
926732	AC1-158 E	85.9551
932781	AC2-102 C	7.8651
932782	AC2-102 E	12.8325
934861	AD1-115 C	4.3569
934862	AD1-115 E	7.1086
941381	AE2-134	7.1599
941831	AE2-188	10.3488
LGEE	LGEE	0.0211
CIN	CIN	0.0710
IPL	IPL	0.0457
G-007	G-007	0.0431
CBM-W2	CBM-W2	0.4716
CBM-W1	CBM-W1	0.1563
MECS	MECS	0.0980
WEC	WEC	0.0180
O-066	O-066	0.2759
CARR	CARR	0.0110
CBM-S1	CBM-S1	0.0580
HAMLET	HAMLET	0.0265
MEC	MEC	0.1155
RENSSELAER	RENSSELAER	0.0090
CATAWBA	CATAWBA	0.0070

16.7 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1948188	926000	AC1-076 TAP	DVP	314778	3PAY TAP	DVP	1	DVP_P7-1: LN 70-2153-B-A	tower	260.0	105.67	114.38	AC	23.35

Bus #	Bus	MW Impact
925671	AC1-043 C	22.1859
925672	AC1-043 E	36.1981
926001	AC1-076 C	17.0287
926002	AC1-076 E	15.9554
926481	AC1-120 C O1	23.1201
926482	AC1-120 E O1	11.9103
926501	AC1-121 C O1	7.9402
926502	AC1-121 E O1	3.7366
932781	AC2-102 C	17.7487
932782	AC2-102 E	28.9585
934861	AD1-115 C	11.0930
934862	AD1-115 E	18.0990
939221	AE1-153 C O1	6.7813
939222	AE1-153 E O1	4.5209
941381	AE2-134	11.7340
941831	AE2-188	23.3536
LGEE	LGEE	0.1019
CIN	CIN	0.3430
CPLE	CPLE	0.1475
IPL	IPL	0.2174
G-007	G-007	0.2341
CBM-W2	CBM-W2	3.8200
CBM-W1	CBM-W1	0.7295
MECS	MECS	0.3430
WEC	WEC	0.0884
O-066	O-066	1.4806
CBM-S2	CBM-S2	0.3657
CARR	CARR	0.0684
CBM-S1	CBM-S1	0.5699
MEC	MEC	0.6699
RENSSELAER	RENSSELAER	0.0540

Affected Systems

17 Affected Systems

17.1 Duke Energy Progress

No Duke Energy Progress impacts were identified as part of this study.

Short Circuit

18 Short Circuit

The following Breakers are overdutied:

None

Stability Analysis

19 Stability Analysis

To be completed during the Facilities Study.

Light Load Analysis

20 Light Load Analysis

No impacts were identified in the Light Load analysis.

Attachment 1

System Configuration

